

Machine Learning - Based Prediction of Semiconductor Device Aging Under Operational Stress

MINI PROJECT REPORT

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UEC2604 MACHINE LEARNING



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EVEN SEM 2025-2026

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BONAFIDE CERTIFICATE

Certified that this mini project titled “**Machine Learning - Based Prediction of Semiconductor Device Aging Under Operational Stress**” is the bonafide work of “**Govindh Subramaniam (3122233002034), Monishraj S (3122233002063) and Moulieswaran (3122233002064)** of VI Semester Electronics and Communication Engineering Branch during Even Semester 2025 – 2026 for UEC 2604 Machine Learning.

Submitted for examination held on _____

INTERNAL EXAMINER

ABSTRACT

Power semiconductor devices, particularly MOSFETs, are critical components in modern electronic systems, and their reliability under prolonged electrical and thermal stress is a significant concern. This work presents a machine learning-based approach for predicting the aging behavior and Remaining Useful Life (RUL) of MOSFETs operating under accelerated thermal overstress conditions. The study utilizes a run-to-failure dataset from the NASA Prognostics Center of Excellence, comprising 105 experiments. Key electrical and thermal parameters, including voltage, current, and temperature, are processed to extract degradation-sensitive features such as on-state resistance $R_{DS(on)}$, which serves as a primary aging indicator.

Multiple machine learning models, including Linear Regression, Polynomial Regression, Random Forest, Gradient Boosting, and Long Short-Term Memory (LSTM) networks, are implemented and evaluated. Traditional regression models exhibit limited performance due to the nonlinear nature of device degradation, whereas ensemble and deep learning methods demonstrate improved predictive capability. Random Forest achieves the lowest prediction error (MAE and RMSE), effectively capturing nonlinear degradation patterns, while Bidirectional LSTM provides the best model fit with superior R^2 by leveraging temporal dependencies in time-series data. The results confirm that semiconductor aging is inherently nonlinear and time-dependent, and data-driven approaches significantly enhance prediction accuracy. The proposed framework supports real-time predictive maintenance and improved reliability in power electronic systems.

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LIST OF SYMBOLS AND ABBREVIATIONS

ABBREVIATION	FULL FORM
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
RUL	Remaining Useful Life
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short-Term Memory
RF	Random Forest
GB	Gradient Boosting
LR	Linear Regression
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
$R_{DS(on)}$	Drain-to-Source On-State Resistance
$dR_{DS(on)}$	Rate of Change of On-State Resistance
V_{ds}	Drain-Source Voltage
I_d	Drain Current
Temp	Temperature
NASA PCoE	NASA Prognostics Center of Excellence
PHM	Prognostics and Health Management

INTRODUCTION

The rapid advancement of machine learning and data-driven modelling techniques has significantly transformed the field of reliability engineering and predictive maintenance. In modern electronic systems, power semiconductor devices such as MOSFETs play a crucial role, and their performance degradation under prolonged electrical and thermal stress poses a major challenge. Accurately understanding and predicting device aging behaviour is essential to ensure system reliability, prevent unexpected failures, and optimize maintenance strategies.

Traditional approaches to reliability assessment primarily rely on physics-based models and empirical regression techniques. While these methods provide useful insights, they often struggle to capture the complex, nonlinear, and time-dependent degradation characteristics exhibited by semiconductor devices under real operating conditions. As a result, their predictive capability for Remaining Useful Life (RUL) estimation is limited, particularly when dealing with large-scale and high-dimensional sensor data.

The emergence of machine learning techniques has introduced a powerful alternative for modeling such complex degradation phenomena. Algorithms such as Linear Regression, Polynomial Regression, Support Vector Machines, and ensemble methods like Random Forest and Gradient Boosting have been applied to learn degradation patterns directly from data. More recently, deep learning architectures, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong capability in modeling temporal dependencies and sequential degradation behavior in time-series data.

This project focuses on analyzing and predicting the aging behavior of power MOSFETs under thermal overstress using machine learning techniques. A run-to-failure dataset from the NASA Prognostics Center of Excellence is utilized, where key electrical and thermal parameters are processed to extract degradation-sensitive features such as on-state resistance $R_{DS(on)}$. Multiple machine learning models are comparatively evaluated for Remaining Useful Life prediction to identify the most effective approach for capturing nonlinear degradation and temporal dynamics. This work bridges traditional reliability methods with data-driven techniques, contributing to predictive maintenance systems for real-time and embedded applications.

LITERATURE SURVEY

Reliability analysis and aging prediction of power semiconductor devices have gained significant attention due to their critical role in modern electronic systems. With the increasing availability of sensor data and run-to-failure datasets, various approaches ranging from physics-based models to advanced machine learning and deep learning techniques have been explored to accurately model degradation behavior and predict Remaining Useful Life (RUL).

1. Semiconductor Device Aging and Failure Mechanisms [1], [2]

A study by S. Mahapatra (2009) highlights the impact of Bias Temperature Instability (BTI) on MOSFET devices. The research demonstrates that prolonged electrical stress leads to threshold voltage (V_{th}) shift, resulting in gradual performance degradation. Similarly, M. Ciappa (2002) analyzed failure mechanisms in power electronic modules and reported that thermal overstress causes progressive parametric degradation before catastrophic failure. These studies establish that semiconductor aging is a gradual and measurable process, making it suitable for predictive modeling using data-driven approaches.

Limitation:

- Physics-based models require detailed domain knowledge and often fail to generalize across different operating conditions.

2. Machine Learning for Aging Prediction [4], [5]

F. Klemme and H. Amrouch (2021) explored the application of machine learning techniques such as Linear Regression, Polynomial Regression, and K-Nearest Neighbors (KNN) for circuit aging estimation under varying workloads. These models learn degradation patterns directly from data without requiring explicit physical modeling.

.Additionally, T. Thekemuriyil (2023) applied machine learning algorithms including Support Vector Machines (SVM), KNN, and Decision Trees to predict aging indicators such as collector-emitter voltage in IGBTs for real-time health monitoring.

Key Contributions:

- Data-driven modeling of degradation
- Ability to capture nonlinear relationships
- Reduced dependency on physical modeling

Limitation:

- Traditional ML models struggle to capture time-dependent degradation trends effectively.

3. Deep Learning for Time-Series Degradation Modeling [3], [6]

Recent advancements in deep learning have enabled more accurate modeling of time-dependent degradation processes. Studies conducted using datasets from the NASA Prognostics Center of Excellence demonstrate the effectiveness of Long Short-Term Memory (LSTM) networks in predicting Remaining Useful Life.

LSTM models are specifically designed for sequential data and can capture long-term dependencies in degradation signals such as on-state resistance (RDS(on)) over time. These models outperform traditional regression approaches by learning temporal patterns and complex nonlinear relationships inherent in semiconductor aging.

Key Findings:

- LSTM effectively models temporal degradation behavior
- Improved prediction accuracy for RUL estimation
- Suitable for run-to-failure and time-series datasets

Limitation:

- Requires large datasets and higher computational resources
- Performance may vary due to device-to-device variability

PROBLEM STATEMENT

Power semiconductor devices, particularly MOSFETs, are widely used in modern electronic systems such as power converters, electric vehicles, and industrial control applications. These devices are continuously subjected to electrical and thermal stress during operation, which leads to gradual degradation over time. If not monitored properly, this degradation can result in unexpected failures, reduced system efficiency, and increased maintenance costs.

Traditional approaches for assessing device reliability are primarily based on physics-based models and offline testing methods. While these methods provide useful insights into failure mechanisms, they are often time-consuming, require extensive domain expertise, and lack the ability to accurately predict the Remaining Useful Life (RUL) under varying real-world operating conditions. Additionally, conventional regression techniques fail to capture the nonlinear and time-dependent nature of semiconductor aging.

With the increasing availability of run-to-failure datasets and sensor measurements, there is a growing need for data-driven approaches that can automatically learn degradation patterns and provide accurate prognostic predictions. However, modeling semiconductor aging is challenging due to complex nonlinear behavior, device-to-device variability, and temporal dependencies in degradation signals.

To address these challenges, this project proposes a machine learning-based framework for predicting the aging behavior and Remaining Useful Life of MOSFET devices under thermal overstress conditions. The system utilizes a dataset obtained from the NASA Prognostics Center of Excellence, where key electrical and thermal parameters are processed to extract degradation-sensitive features such as on-state resistance $R_{DS(on)}$.

Multiple machine learning models, including Linear Regression, Polynomial Regression, Random Forest, Gradient Boosting, and LSTM networks, are implemented and evaluated to identify the most effective approach for RUL prediction. The proposed method captures nonlinear degradation behavior and temporal patterns in aging data, enabling early failure detection, improved reliability, reduced downtime, and support for real-time predictive maintenance in embedded and industrial applications.

METHODOLOGY AND MODEL DEVELOPMENT

4.1 Data Collection & Preprocessing

The dataset used in this project is obtained from the NASA Prognostics Center of Excellence], which provides run-to-failure data for power MOSFET devices under thermal overstress conditions. The dataset is available in MATLAB (.mat) format, where each file represents a complete degradation cycle of a device from its healthy state to failure.

A total of **106 .mat files** are collected, corresponding to **106 device cycles**. The data loading process is performed programmatically and completed in approximately **292.61 seconds**. Each file contains raw measurements of electrical and thermal parameters such as drain-source voltage, drain current, and package temperature.

Table 4.1 Dataset Features

FEATURES	DESCRIPTION
Time Index	Represents the progression of the aging cycle
Voltage	Drain-source voltage measurements
Current	Drain Current Values
Temperature	Device operating temperature
R _{DS(on)}	On-state resistance (computed aging indicator)
dR _{DS(on)}	Rate of change of resistance
RUL	Remaining Useful Life (target variable)

The dataset is preprocessed by extracting steady-state measurements and removing noise and inconsistencies in the raw signals. Since the dataset does not directly provide on-state resistance, it is computed using voltage and current values, which serves as a key degradation indicator.

Preprocessing steps include:

- Removal of noisy and unstable measurements
- Smoothing of signals to reduce fluctuations
- Normalization of features using standard scaling
- Handling missing or inconsistent values

After preprocessing, the dataset is structured into a feature matrix of shape **(106, 28)**, indicating 106 samples with 28 extracted features per cycle. The corresponding target vector (RUL) has a shape of **(106,)**.

The dataset is then split into training and testing sets using an 80:20 ratio, resulting in:

- **Training samples:** 84
- **Testing samples:** 22

These preprocessing steps transform raw sensor data into structured numerical inputs suitable for machine learning and deep learning models.

4.2 Feature Extraction

Feature extraction is a crucial step in this project, where raw electrical and thermal measurements are transformed into meaningful parameters that reflect device degradation. Since raw signals alone do not directly indicate aging behavior, domain-specific features are derived to improve prediction accuracy.

In this work, multiple feature extraction techniques are applied to capture both static and dynamic characteristics of semiconductor degradation.

4.2.1 Feature Representation for Machine Learning Models

For classical machine learning models such as Linear Regression, Polynomial Regression, Random Forest, and Gradient Boosting, the extracted features are used in structured numerical form. The raw MOSFET data consists of electrical and thermal measurements, which are transformed into meaningful degradation indicators during preprocessing.

These features include voltage, current, temperature, and the computed on-state resistance $R_{DS(on)}$ along with its rate of change. Feature representation focuses on scaling and

normalization rather than vectorization. Standardization techniques such as **StandardScaler** are applied to ensure that all features have a uniform scale, improving model convergence and performance.

This structured feature representation allows traditional machine learning models to effectively capture nonlinear relationships between input variables and Remaining Useful Life (RUL).

4.2.2 Time-Series Sequence Representation for LSTM

For the Long Short-Term Memory (LSTM) model, the data is treated as a time-series sequence to capture the temporal nature of MOSFET degradation.

A sliding window approach is used to convert the dataset into sequences, where each input sample consists of a sequence of past observations (e.g., resistance, temperature, and other features over time). This enables the model to learn how degradation evolves across cycles.

Since sequence lengths must be consistent, the data is reshaped into fixed-size windows before being fed into the LSTM model. The features are normalized prior to sequence generation to ensure stable training.

Unlike traditional models, LSTM networks are capable of learning long-term dependencies and time-dependent patterns in degradation data. This makes them particularly suitable for predicting Remaining Useful Life (RUL), where past behavior strongly influences future degradation.

4.2.3 Degradation Feature Engineering and Temporal Modeling

In this project, feature engineering plays a critical role in improving prediction accuracy by extracting meaningful indicators of device aging.

Key engineered features include:

- **On-state resistance (RDS(on)):** Primary aging indicator that increases over time
- **Resistance gradient (dRDS(on)):** Captures the rate of degradation
- **Temperature influence:** Reflects thermal stress effects on aging
- **Time index / cycle progression:** Represents degradation timeline

These features are designed to capture:

- **Physical degradation behavior** of MOSFET devices
- **Nonlinear relationships** between operating conditions and aging
- **Temporal trends** in degradation patterns

By combining domain-specific feature engineering with machine learning and deep learning models, the system effectively models semiconductor aging and improves the accuracy of Remaining Useful Life prediction.

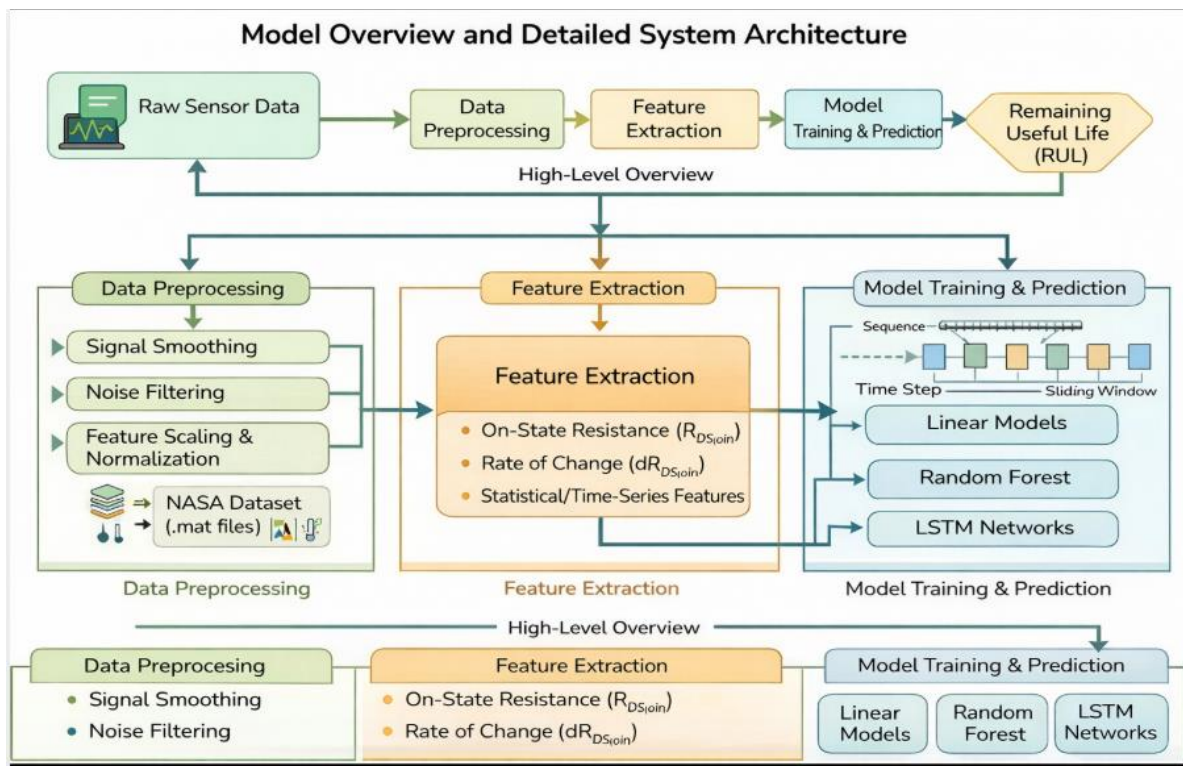


Figure 4.2.3.1 - Detailed Architecture Diagram

4.3 Model Development

The model development phase focuses on designing, implementing, and evaluating multiple machine learning and deep learning models to predict the Remaining Useful Life (RUL) of semiconductor devices under operational stress. The objective is to capture both nonlinear degradation behavior and temporal dependencies present in the MOSFET aging dataset. To ensure a comprehensive analysis, a combination of traditional regression models, ensemble learning methods, and sequence-based deep learning models were developed.

4.3.1 Regression-Based Models

Linear Regression

Linear Regression was implemented as a baseline model to establish a simple relationship between input features and RUL. It assumes a linear dependency between degradation parameters and device life. The simplest model assumes the RUL denoted as y ,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \quad \text{..... (1)}$$

Where:

- β_0 : Intercept.
- β_i : Coefficients for each degradation parameter.
- ϵ : Error term.

However, MOSFET aging is inherently nonlinear, and hence this model showed limited predictive capability.

Polynomial Regression

To address nonlinearity, Polynomial Regression was applied by introducing higher-order feature interactions. This model improves flexibility compared to linear regression but still struggles to generalize complex degradation patterns over time. To account for the nonlinear aging of MOSFETs, features are raised to the power of d :

$$y = \beta_0 + \sum_{i=1}^n \beta_i x_i + \sum_{i=1}^n \beta_{ii} x_i^2 + \cdots + \epsilon \quad \text{..... (2)}$$

4.3.2 Ensemble Learning Models

Random Forest Regression

Random Forest is an ensemble learning technique that constructs multiple decision trees and aggregates their outputs. It effectively captures nonlinear relationships and reduces overfitting through averaging. Random Forest is an ensemble of B decision trees. The final prediction is the average of all individual tree predictions $h_b(x)$:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B h_b(x) \dots\dots\dots (3)$$

This model demonstrated strong performance due to:

- Ability to model complex feature interactions
- Robustness to noise in experimental data
- Improved generalization across different stress conditions

Gradient Boosting Regression

Gradient Boosting builds models sequentially by minimizing prediction errors of previous models. It focuses on difficult samples and incrementally improves performance. This model builds sequentially. If $F_m(x)$ is the model at iteration m , the next iteration adds an estimator $h_m(x)$ to minimize the loss L :

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \dots\dots\dots (4)$$

Where γ in equation (4) is the step size (learning rate) and $h_m(x)$ is fit to the negative gradient (residuals) of the loss function. Although it captures nonlinearities effectively, its performance depends heavily on hyperparameter tuning and may be prone to overfitting if not carefully optimized.

4.3.3 Sequence-Based Deep Learning Model

Long Short-Term Memory (LSTM)

To capture the time-dependent nature of device degradation, a stacked LSTM model was developed. Unlike traditional models, LSTM can learn temporal dependencies and sequential patterns in aging data. The model development involved the following steps:

- Generation of time-series sequences using a sliding window approach
- Normalization of input features
- Training a stacked LSTM network on sequential data
- Prediction of RUL for unseen test sequences

The LSTM model effectively learns degradation trends over time but may require large datasets and careful tuning for optimal performance.

4.3.4 Bidirectional LSTM with Attention Mechanism (BiLSTM + Attention)

To further enhance temporal learning, a Bidirectional LSTM (BiLSTM) combined with an Attention mechanism was implemented. Unlike standard LSTM, BiLSTM processes the input sequence in both forward and backward directions, enabling the model to capture past and future context simultaneously.

The addition of the Attention mechanism allows the model to:

- Assign different importance weights to different time steps
- Focus on critical regions of the degradation sequence
- Improve interpretability of the model's predictions

4.4 Model Training and Evaluation

All models were trained and evaluated using standard regression performance metrics:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination (R2)

The evaluation process ensures a fair comparison across models and helps identify the most suitable approach for RUL prediction.

Table 4.3.4.1 Linear regression evaluation results

R2 Score(%)	MAE(%)	RMSE(%)
0.76	26.42	32.11

Table 4.3.4.2 Polynomial regression evaluation results

R2 Score(%)	MAE(%)	RMSE(%)
11.39	28.29	32.45

Table 4.3.4.3 Random forest evaluation results

R2 Score(%)	MAE(%)	RMSE(%)
68.6	2.48	4.75

Table 4.3.4.4 Gradient boosting results

R2 Score(%)	MAE(%)	RMSE(%)
73.13	12.12	14.54

Table 4.3.4.5 LSTM results

R2 Score(%)	MAE(%)	RMSE(%)
90.61	1088.35	1952.53

Table 4.3.4.6 BiLSTM+attention results

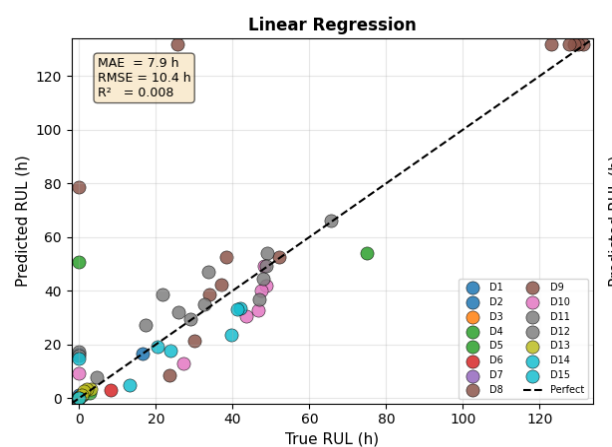
R2 Score(%)	MAE(%)	RMSE(%)
98.87	48.82	69.16

OBSERVATIONS AND RESULTS

5.1 Model Performance Results

In this project, multiple models including Linear Regression, Polynomial Regression, Random Forest, Gradient Boosting, LSTM, and BiLSTM with Attention were implemented for predicting the Remaining Useful Life (RUL) of MOSFET devices.

The performance of each model was evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R2). Among all models, BiLSTM+attention achieved the best performance, capturing temporal dependencies in degradation data, providing accurate predictions and strong generalization.

**Fig 5.1.1 Linear Regression model**

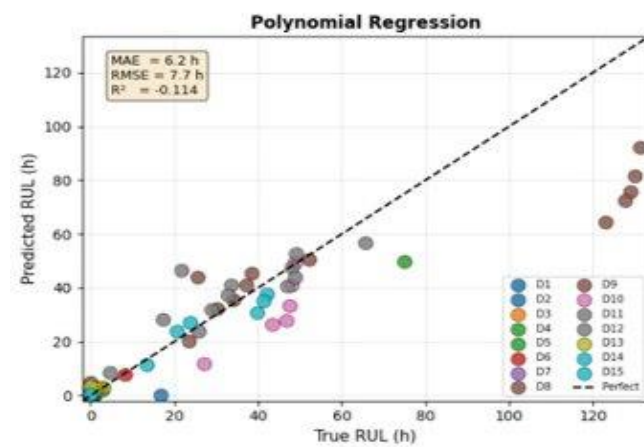


Fig 5.1.2 Polynomial Regression model

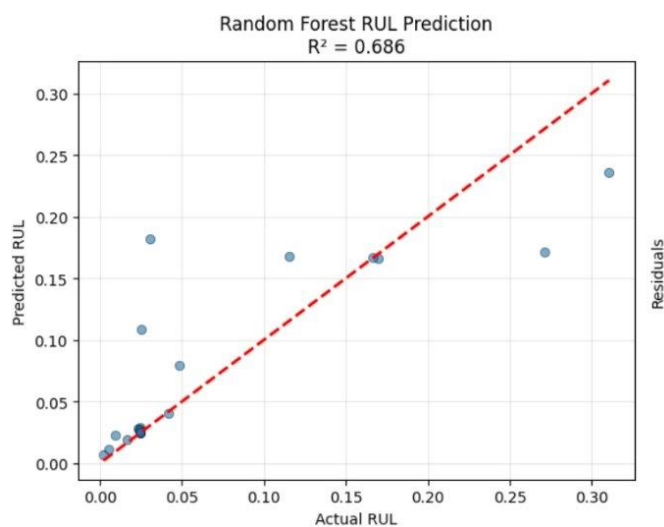


Fig 5.1.3 Random Forest Model

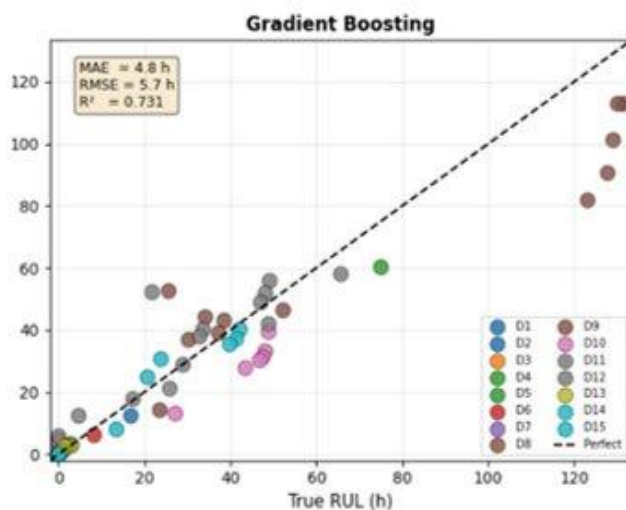


Fig 5.1.4 Gradient boosting model

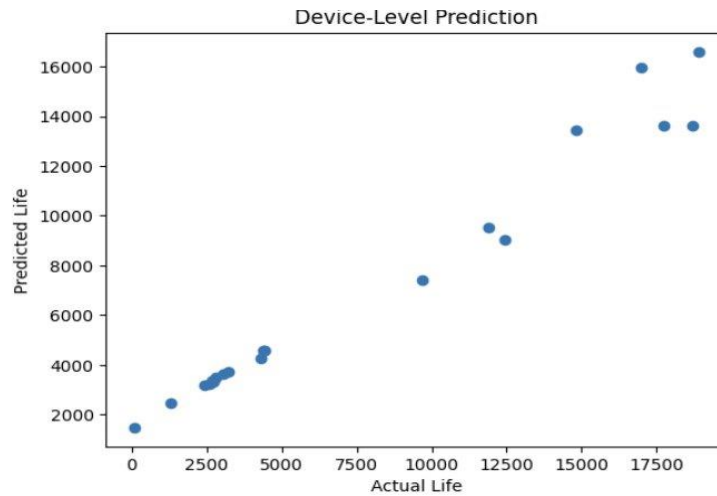


Fig 5.1.5 LSTM model

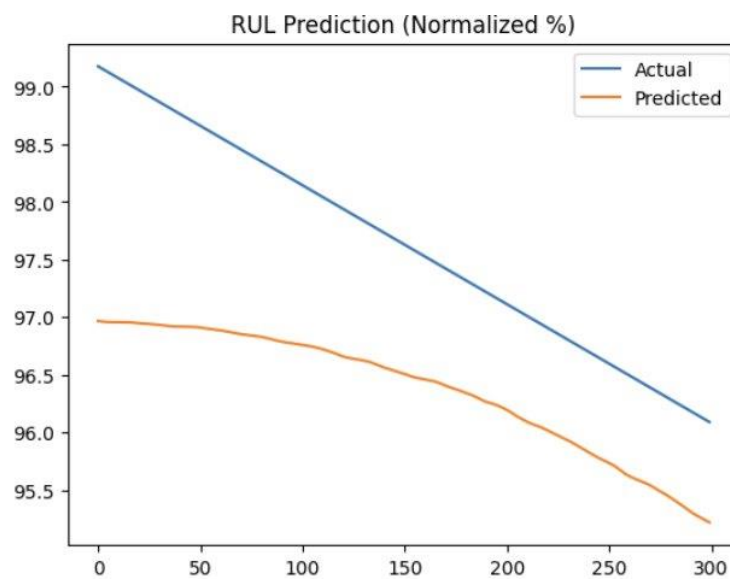


Fig 5.1.6 BiLSTM + attention model

5.2 Observations

The following observations were made from the experimental results:

- Semiconductor device aging exhibits nonlinear and time-dependent behavior.
- Linear and Polynomial Regression models showed poor accuracy due to their limited modeling capability.
- Random Forest effectively captured nonlinear relationships and performed consistently well.
- LSTM improved prediction by learning sequential degradation patterns.

- BiLSTM with Attention further enhanced performance by focusing on important time steps.
- Model performance improves with better feature extraction and data quality

5.3 RUL Prediction Analysis

The BiLSTM with Attention model achieved the best evaluation metrics, indicating superior predictive performance. The Random Forest model also produced predictions closely aligned with actual RUL values, showing strong generalization.

The LSTM and BiLSTM + Attention models effectively captured the degradation trend, although slight deviations were observed in later stages of device life. Overall, the prediction curves follow the actual degradation pattern, demonstrating successful learning of aging behavior.

MOSFET RUL Predictor
TSK-based remaining useful life estimator — enter electrical measurements below

Electrical inputs

Supply voltage V 28	Drain-source voltage V 0.8
Drain current A 5.0	Flange temperature °C 85

Degradation trend assumptions

V_{ds} drift rate		+0.0005 V/cyc
I_d drift rate		-0.0050 A/cyc
T_{flange} drift rate		+0.05 °C/cyc

Estimate RUL →

17,942,343 Estimated RUL (s)	4984.0 Hours remaining	Device health Healthy
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Fig 5.3.1 Healthy MOSFET Prediction

MOSFET RUL Predictor
TSK-based remaining useful life estimator — enter electrical measurements below

Electrical inputs

Supply voltage V 28	Drain-source voltage V 3.5
Drain current A 0.8	Flange temperature °C 145

Degradation trend assumptions

V_{ds} drift rate	<input type="range"/>	+0.0005 V/cyc
I_d drift rate	<input type="range"/>	-0.0050 A/cyc
T_{flange} drift rate	<input type="range"/>	+0.05 °C/cyc

Estimate RUL →

9,654,207 **2681.7**

Estimated RUL (s) Hours remaining

Device health

Degrading

Fig 5.3.2 Degrading MOSFET Prediction

CONCLUSION AND FUTURE WORK

In conclusion, this project demonstrates the effective use of machine learning and deep learning techniques for predicting the aging behavior and Remaining Useful Life (RUL) of power MOSFET devices. Using run-to-failure data from the NASA Prognostics Center of Excellence, the system successfully models degradation under thermal overstress conditions. Comparative analysis shows that Linear and Polynomial Regression are less effective for capturing nonlinear aging behavior, while Random Forest and Gradient Boosting improve predictive performance. Among all models, BiLSTM achieves the highest accuracy, while LSTM effectively captures temporal degradation patterns.

The results confirm that semiconductor aging is nonlinear and time-dependent, and that data-driven approaches significantly improve prediction accuracy over conventional methods. The developed system offers a practical framework for predictive maintenance, supporting early fault detection and reducing unexpected failures.

Overall, this project provides a scalable and efficient solution that can be further improved through advanced hyperparameter tuning, validation on diverse MOSFET datasets, and hybrid physics-informed machine learning models. Future work may also focus on improving interpretability, reliability, and the inclusion of additional environmental and operational parameters.

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